BHASHALENS

MODEL CARD · Indian Language Identification Using Character-Level NLP

Course: Natural Language Processing (NLP) CA3 **Model:** Character TF-IDF + Logistic Regression

1. Model overview and intended use

BhashaLens predicts which of 18 known language classes best matches an input headline. It is an educational prototype for preliminary language identification, not a definitive system for every context. It predicts only among classes represented in its training data.

2. Dataset

Dataset: Indian Language Identification (downloaded from Kaggle). The notebook reported 95,045 training examples, 20,368 validation examples, and 20,368 test examples after removing rows with missing text or labels. Text and labels were read from the **Headline** and **Language** columns. The notebook reported 18 language classes.

3. Method

The pipeline uses character-level TF-IDF features with character n-grams of length 2–5, followed by Logistic Regression (maximum 300 iterations, random seed 42). Missing text/label rows were removed. The training split was used to fit the model, the validation split for checking, and the test split for final evaluation.

4. Test performance

Metric	Result
Test accuracy	99.73% (0.997299)
Test macro F1-score	99.68% (0.996752)
Absolute short-vs-long accuracy gap	0.114 percentage points

5. Performance audit

The first audit compared short and long headlines using the median word count of the training data as the threshold. The reported absolute accuracy gap was 0.114 percentage points. This small observed gap alone does not prove equivalent performance; subgroup sample counts and scores should be reviewed. The second audit reports per-language recall/F1 across the 18 classes in the notebook's language-wise audit output.

6. Explainability

Logistic Regression coefficients were inspected to identify character n-grams associated with each language class. These are learned associations, not proof that a character sequence causes a prediction. The notebook includes top patterns for selected languages and a confusion matrix to inspect confusions.

7. Limitations and responsible use

Performance may be lower for very short text, mixed-language content, transliterated text, noisy spelling, or languages not represented in training. The dataset may not represent all real-world writing contexts. High test performance may reflect distinctive scripts and dataset characteristics. Use predictions as preliminary assistance, not as authoritative decisions.

Before submission: add student name/PRN if required, verify the dataset licence/source details, and confirm the topic was approved. Expand audit wording if your instructor expects exact per-group counts and metrics.